

Speech — 30 Minutes

Exploring Excess Return Generation through European Fixed Income Arbitrages

PhD Research Presentation Series — June 2026

Opening (slides 1–4)

Slide 1 — Title page

≈19s · elapsed ≈0:19

[Slide 1] Title

Good afternoon, everyone — thanks for being here. I'm going to present the first paper of my dissertation titled *Exploring Excess Return Generation through European Fixed Income Arbitrages*.

[Brief pause. Move to slide 2.]

Slide 2 — Motivation: The Steamroller Question

≈89s · elapsed ≈1:48

[Slide 2] Motivation

So let me start with the question behind the paper. Back in 2007, Duarte, Longstaff and Yu asked whether fixed-income arbitrage earns genuine alpha, or whether the returns are just pay for systematic risk — what they called picking up nickels in front of a steamroller. They found that on U.S. data, the more complex strategies did earn alpha that survived a ten-factor benchmark model. We take that question to European markets, and to three strategies with a *deterministic* payoff at maturity: if they can be held to maturity, the profit is certain, whatever markets do in between. Duarte's strategies don't have this property. In fact, their payoffs depend on realized volatility, prepayment, and so on, so the outcome is never certain. The three strategies — also traded on the Prop Trading Desk where I work — are the BTP Italia inflation-linked basis, the CDS–Bond basis, and the CDS index skew.

[Move to slide 3.]

Slide 3 — Research Questions

≈47s · elapsed ≈2:35

[Slide 3] Research Questions

The paper is built around three sets of questions, one per section. First: do these strategies generate alpha that survives standard benchmark factor models? Second: does that alpha survive a far more demanding battery of controls — and is it state-dependent, moving with intermediary funding stress? And finally, the most important one: if the alpha is there, why does it persist — do the three mispricings move together, the way the slow-moving capital literature predicts?

Slide 4 — Novelty and Contribution

≈84s · elapsed ≈3:59

[Slide 4] Novelty and Contribution

Our contribution lies in four respects. First, the strategies and the data: to the best of our knowledge these three have never been studied jointly as systematic arbitrages, and the BTP Italia basis has had *no* prior academic study at all.

Second, the payoff: all three converge to a *certain* profit if held to maturity, unlike the trades Duarte studied.

Third, a data-driven approach to risk adjustment: instead of a fixed benchmark, we apply rolling PCA and machine-learning selection over eighty-five candidate factors from the literature, letting the data tell us which ones price these returns.

Fourth, and most importantly, ((we do not stop at documenting alpha.)) We test *why* it persists, by mapping our findings to the predictions of the slow-moving capital literature and by exploiting holder-base heterogeneity in the European setting as the source of identification.

The Three Strategies (slides 5–7)

Slide 5 — Strategy 1 — BTP Italia: The Inflation-Linked Basis $\approx 102s$ · elapsed $\approx 5:41$

[Slide 5] Stra

Let me start with the BTP Italia — an Italian government bond linked to domestic inflation. Following Fleckenstein, Longstaff, and Lustig (2014), the basis is defined as the differential between the total yield of the inflation-linked bond — converted to fixed via an inflation ASW — and the yield of the maturity-matched nominal BTP.

The TIPS comparison is telling: the basis was large before 2010 — with peak of about 200 basis points during the financial crisis — but has effectively vanished as institutional capital moved in. The BTP Italia basis, by contrast, *persists*.

Two features distinguish the BTP Italia from other IL bonds. First, there is no ballooning risk: the inflation revaluation is paid out semiannually, so the accumulated revaluation never contaminates default exposure. Second, the bondholder base is principally *retail*: primary placement is reserved to households for the first days of issuance, with a loyalty bonus for holding to maturity. So they are preferred-habitat investors in the sense of Vayanos and Vila — they use no leverage, face no margin calls, and are not forced to sell in stress.

Slide 6 — Strategy 2 — CDS–Bond Basis $\approx 58s$ · elapsed $\approx 6:39$

$\approx 58s$ · elapsed $\approx 6:39$

[Slide 6] Stra

The second strategy is the negative CDS–Bond basis, defined as the difference between an issuer’s CDS spread and the z-spread of one of its bonds, maturity-matched. When the basis is negative, you buy the bond, buy CDS protection and also hedge the interest- rate risk, locking in positive carry until convergence or maturity. And the trade is default-immune: in a default, bond plus protection return par regardless of recovery.

((The universe is restricted to single-name CDS in the on-the-run iTraxx Main — which ensures liquidity and investment-grade quality — and all are centrally cleared, so there is no counterparty risk.))

Slide 7 — Strategy 3 — iTraxx Combined: The CDS Index Skew $\approx 80s$ · elapsed $\approx 7:59$

[Slide 7] Stra

The third strategy is the CDS index skew, defined as the difference between the CDS index traded spread and its fair spread implied by the single-name constituents.

Three forces keep that skew from closing. First, the liquidity differential: single-name CDS are far less liquid than the index, especially under stress— and the skew is a well-known marketwide liquidity proxy. Second, the institutional investors use the index as the natural macro credit hedge,

generating concentrated demand that pushes the index leg tighter. Third, replicating the basket of up to one hundred and twenty-five names against one index contract is costly.

((Our strategy trades the four on-the-run European iTraxx sub-indices, with both legs under a single netting set, without counterparty risk. Note this is a pure derivative trade — no cash bond to finance.))

Construction and Performance (slides 8–12)

Slide 8 — Market-Level Basis Time Series

≈19s · elapsed ≈8:18

[Slide 8] Mar

Here are the three raw mispricings over time. Notice that they remain chronically open and do not share the same dynamics.

Slide 9 — Trade Construction and Index Aggregation

≈61s · elapsed ≈9:19

[Slide 9] Tra

Following Duarte and co-authors, each trade is treated as a separate little fund, and the strategy index is the weighted average across the funds that are live.

We consider two weighting schemes: the equal-weighted, as in Duarte, and the signal-weighted, from Rebonato and Ronzani (2021), where a bigger dislocation gets a bigger allocation. Signal-weighting dominates on both Sharpe and the manipulation-proof measure, so we adopt it as the baseline; equal-weighted results are in the appendix. And throughout, we charge transaction costs that rise with market stress, as they do in practice — so costs bite hardest exactly when returns are biggest.

Slide 10 — Strategy Performance — Summary Statistics

≈34s · elapsed ≈9:53

[Slide 10] Str

All three strategies generate positive, economically significant excess returns, net of those costs. The Sharpe ratios run from about 0.7 on the BTP Italia and the iTraxx skew up to 1.7 on the CDS–Bond basis, ((which is large for fixed income strategies.)) And against the steamroller story, all three show *positive* return skewness, with moderate drawdowns.

Slide 11 — Cumulative Returns and Drawdowns

≈15s · elapsed ≈10:08

[Slide 11] Cu

((This is just the cumulative return of the three strategies: they all rise steadily, with shallow dips that bounce back fast.))

Slide 12 — MPPM and Volatility-Managed Performance

≈48s · elapsed ≈10:56

[Slide 12] M

Two diagnostics complete the picture. The manipulation-proof performance measure of Goetzmann and co-authors; we see that it's almost *invariant* to risk aversion, so the Sharpe ratios aren't hiding tail risk. And the Moreira and Muir volatility-managed test. It asks whether each strategy's risk-reward holds steady over time. For the BTP and the skew, it does. For the CDS–Bond, it moves — and that's the only trade that isn't cash-neutral. That's our first hint of the regime story.

Benchmark Factor Models (slides 13–17)

Slide 13 — Factor Frameworks for Performance Attribution $\approx 39s$ · *elapsed $\approx 11:35$*

[Slide 13] Fa

Having documented positive excess returns, now the question is: do they survive standard risk controls, or are they just pay for systematic risk?

We run three canonical benchmark models, each one twice — with the original U.S. factors, and with euro counterparts, since these are European strategies. ((The point of running three is to be sure no single factor set drives the result.))

Slide 14 — Duarte — 10-Factor Model $\approx 23s$ · *elapsed $\approx 11:58$*

[Slide 14] Du

The first specification is Duarte’s ten-factor model built for fixed-income arbitrage, testing for equity, corporate bond, and Treasury exposures. We see that the alpha is positive and significant at one percent for all three strategies.

Slide 15 — Fung–Hsieh — 7-Factor Hedge Fund Model $\approx 25s$ · *elapsed $\approx 12:23$*

[Slide 15] Fu

The second specification is Fung–Hsieh 7 factor asset-based model that tests whether hedge fund returns come from equity, fixed-income and trend-following factors. The picture is the same: the alphas remain positive and significant, in the same range.

Slide 16 — Brooks et al. — Active Fixed Income Premia $\approx 29s$ · *elapsed $\approx 12:52$*

[Slide 16] Br

The third specification is Brooks and co-authors, that decompose the returns of active fixed-income managers into 8 traditional risk premia (duration, credit, volatility, emerging-market, inflation-linked, and so on.). Again, all three alphas remain positive and significant at one percent ((—though this specification produces fewer significant loadings.))

Slide 17 — Cross-Model Synthesis $\approx 66s$ · *elapsed $\approx 13:58$*

[Slide 17] Cr

Three things bring the section together. First, the full-sample alpha is consistent across all three frameworks — positive and significant for every strategy, with stable magnitudes despite the different factor structures.

Second, when we split by stress regime, the HIGH-stress alpha is between two and four times the low-stress alpha — in every cell of the table, and across all three benchmarks. Third, the conditional specification makes this continuous: we let the alpha itself vary with stress, and the slope is positive everywhere, significant at one percent for the two institutional strategies. The implication is: alpha rises systematically when intermediary capital is scarce — precisely the signature of slow-moving capital that Duffie identifies.

Factor Selection: PCA and AEN (slides 18–26)

Slide 18 — Why Go Beyond Standard Benchmarks? $\approx 42s$ · *elapsed $\approx 14:40$*

[Slide 18] W

Now, the three benchmarks give us positive, significant alpha — but their adjusted R-squared runs only from one to eighteen percent, ((so most of the return variation is left unexplained.)) That raises the question: are there *other* factors, outside the standard benchmarks, that span these returns? So after an extensive literature review, we assemble eighty-five candidate factors and apply two methods to find the ones that matter.

Slide 19 — Systematic Extension of Risk Adjustment

≈25 s · elapsed ≈15:05

[Slide 19] Sy

These factors span six economic families. But the point is the number at the bottom: with 85 factors, the number of possible models is two-to-the-eighty-five. Far too many to search by hand, which is why we need a penalised method with a formal selection guarantee.

Slide 20 — PCA — Implementation and Spanning Regression

≈39 s · elapsed ≈15:44

[Slide 20] PC

We start with a rolling principal component analysis, following Ludvigson and Ng (2009).

Two results emerge. Alpha survives at one percent for all three strategies, under both timings — contemporaneous and predictive. But the adjusted R-squared remains low, from zero to twelve percent. The diffuse principal components do not span the returns, which motivates the move to a supervised, sparse approach.

Slide 21 — Adaptive Elastic Net — Methodology

≈33 s · elapsed ≈16:17

[Slide 21] Ad

Our main tool is the Adaptive Elastic Net of Zou and Zhang — a penalised, machine-learning method that picks a few factors from the eighty-five while handling their collinearity, with a formal selection guarantee. We harden it with the stability selection of Meinshausen and Bühlmann, and estimate the alpha on that stable set.

Slide 22 — AEN Stable Set — BTP Italia

≈26 s · elapsed ≈16:43

[Slide 22] AE

For the BTP Italia, six factors survive: ((the Amihud illiquidity measure, the Libor–OIS spread, the ten-year swap spread, plus two equity style factors.)) So the method picked liquidity, funding, term structure, and equity factors. And the alpha survives, significant at one percent.

Slide 23 — AEN Stable Set — CDS–Bond Basis

≈54 s · elapsed ≈17:37

[Slide 23] AE

The CDS–Bond basis is the clean case. Its eight survivors map almost perfectly onto the intermediary channel: the European prime-broker CDS spread, which is literally dealer counterparty risk, ((plus financial uncertainty, US credit spread, the option-implied equity premium — the SVIX bound of Professor Ian Martin, which I picked up during his elective here at EDHEC last January —, the five-year swap spread, and the excess return on industrial corporate bond.)) Just what slow-moving capital points to for a trade that finances a cash bond on a dealer's balance sheet. The alpha holds here too, at one percent.

Slide 24 — AEN Stable Set — iTraxx Combined

≈45 s · elapsed ≈18:22

[Slide 24] AEN Stable Set — iTraxx Combined

The

iTraxx skew is the exception, and the informative one. Its five factors are all aggregate funding-stress and uncertainty proxies: ((the option-implied equity premium, Libor–OIS, European equity

volatility, real uncertainty, and U.S. Treasury settlement fails.)) That fits its role as a market-wide liquidity proxy. But here the alpha collapses and is no longer significant. The factors swallow the whole return, so the skew's premium is simply pay for systematic liquidity risk.

Slide 25 — Cross-Method Comparison

≈49 s · elapsed ≈19:11

[Slide 25] Cr

((This table lines up every method side by side.)) We see that the adjusted R-squared jumps from the range of one to eighteen percent under classical benchmarks to approximately twenty-two to twenty-seven - ((the selected factors absorb far more of the return variation.)) And it answers our second question: the alpha survives both PCA and the AEN for the BTP Italia and the CDS–Bond basis, while the iTraxx skew is fully absorbed — itself an informative exception.

Slide 26 — AEN-Selected Factors — Economic Reading

≈25 s · elapsed ≈19:36

[Slide 26] AE

The factors are economically interpretable, and each strategy loads on a different mix. That's the argument: there's no single risk common to all three, so the surviving alpha can't be compensation for one. Each carries its own.

Cross-Strategy Interdependencies (slides 27–36)

Slide 27 — Cross-Strategy Interdependencies — Setup

≈29 s · elapsed ≈20:05

[Slide 27] Cr

((So we've established the alpha is real, and that it strengthens in stress.)) The deeper question is why it persists, and whether the three mispricings are linked. We anchor this in the slow-moving capital literature: ((the idea that intermediary capital moves across markets only slowly.))

Slide 28 — Five Testable Predictions of Slow-Moving Capital

≈18 s · elapsed ≈20:23

[Slide 28] Fiv

From this theory we draw five testable predictions: co-movement, amplification under stress, lead-lag transmission, persistence, and a common driver. We take them one at a time.

Slide 29 — Investor-Base Segmentation — A Source of Identification

≈21:48

≈85 s · elapsed

[Slide 29] Inv

Before the tests, a key point about identification. All three strategies are run by the same kind of arbitrageur — hedge funds and prop desks, for example. What differs is who holds the instruments. The CDS–Bond basis and the iTraxx skew are held by institutions, with margin calls and VaR limits, so when stress hits, their risk budget blows up, and they are forced to unwind at once. That's the margin-spiral mechanism of Brunnermeier and Pedersen. The BTP Italia, by contrast, is held by the retail, the preferred-habitat investors we met before, who don't need to unwind during stress.

This heterogeneity is the source of identification. If co-movement were driven by macro news, all three would move together regardless of who holds them. Under slow-moving capital, it instead depends on sharing a capital pool — so the sign of every correlation turns on whether the pool is shared.

Slide 30 — P1 — Cross-Market Correlation

≈61 s · elapsed ≈22:49

[Slide 30] P1

The first prediction says that mispricings serviced by a common pool of intermediary capital should co-move. ((We test it with pairwise correlations of monthly mispricing changes.)) The prediction holds for the institutional pair: CDS–Bond and iTraxx — the correlation is positive, significant at ten percent. The retail-held BTP Italia goes the other way: against the CDS–Bond basis it’s negative, and strongly so, at one percent. That doesn’t break P1 — which predicts co-movement only for trades sharing the pool, and the retail BTP Italia doesn’t. ((So that negative sign is the first empirical signal of the segmentation we exploit for identification.))

Slide 31 — P2 — Stress Amplification

≈91 s · elapsed ≈24:20

[Slide 31] P2

The second prediction sharpens the first: the co-movement should get stronger when dealer capital is scarce. ((We test it two ways: with a regime split, and with an interaction regression with stress entering continuously.)) For the institutional pair, the correlation climbs from mildly negative in calm markets to clearly positive in stress, significant at one percent on a Fisher z-test, and a Forbes–Rigobon correction confirms this is not just a volatility artefact. The interaction term points the same way and is significant at ten percent; so the prediction holds for the institutional pair on both tests.

For the BTP Italia the sign flips: under stress its correlation with the CDS–Bond basis deepens to clearly negative, significant at five percent, with the interaction term confirming at one percent. ((So stress doesn’t amplify a positive link here; it deepens a negative one.)) That’s the mirror image of the institutional pattern, exactly what segmentation predicts.

Slide 32 — P3 — Lead-Lag Transmission

≈70 s · elapsed ≈25:30

[Slide 32] P3

Prediction three says that, if capital moves slowly across markets, then a change in one mispricing should help predict another, running from the more liquid market to the less liquid.

Two things line up. The spanning regression shows genuine lead-lag predictability: lagged iTraxx changes predict the CDS–Bond basis, and the CDS–Bond basis in turn predicts BTP Italia.

And a Granger test fixes the direction: the iTraxx index Granger-causes both other bases at one percent, and neither reverses. Thus shocks originate in the most liquid market, where dealer inventory adjusts fastest, and reach the less liquid bases within a month—exactly the timescale Mitchell and Pulvino document.

Slide 33 — P4 — Persistence Reflects Funding-Dependence

≈52 s · elapsed ≈26:22

[Slide 33] P4

Prediction four links persistence to funding-dependence: The more a trade leans on financing a cash position, the longer a shock takes to fade. We fit an AR(1) on each mispricing level and read off the half-life. The CDS–Bond basis is the most persistent—about three months—and it’s the only one that is not cash-neutral. The other two bounce back faster, around two months. The theory predicts a full ordering by funding-dependence, so that makes P4 *partially* confirmed: ((it holds at the funding-heavy end, but the two less funding-intensive strategies come out essentially identical.))

Slide 34 — P5 — Common Factor: Intermediary Capital, Not Macro

≈27:29

≈67 s · elapsed

[Slide 34] P5

P5 concerns the source of the co-movement. If slow-moving intermediary capital is the binding constraint, then the common factor behind the three mispricings should load on intermediary balance-sheet variables, not on macroeconomic ones. So we take the first principal component of the three mispricing-change series and race 2 sets of regressors against it. The first is the intermediary set ((— made up of capital ratios, funding spreads, illiquidity measure.)) Every one comes in with the predicted sign and together they explain about a quarter of the common factor. The second is a macroeconomic placebo, ((made up of macro, real-activity, and policy uncertainty, plus long-term inflation expectations.)) This set explains roughly half as much, and not one of its variables is significant. ((The contrast confirms that the common factor is intermediary balance-sheet, not macroeconomic.)) P5 is confirmed.

Slide 35 — All Five Predictions — Scorecard

≈25s · elapsed ≈27:54

[Slide 35] All

The scorecard collects the verdicts. Four of the five predictions find unambiguous support in the data, while P4—the persistence ordering—is partly there: ((only the CDS–Bond basis stands apart, while the other two look essentially identical.))

Slide 36 — Why Do These Arbitrages Persist? (close)

≈96s · elapsed ≈29:30

[Slide 36] W

Let me finish where I started: why haven't these trades been competed away? Three forces, reinforcing each other.

First, regulation: Boyarchenko and co-authors show that post-crisis leverage rules permanently raised the cost of holding a basis trade on a dealer's balance sheet — and we see it in how persistent the CDS–Bond basis is. Second, slow-moving capital: capital moves across the dealer network only over months. We see that the iTraxx adjusts first, and the BTP Italia last. And third, segmentation: the BTP Italia's retail holders are a preferred habitat, insulated from institutional unwinding, so its basis stays open.

So, back to Duarte. For two of the three — BTP Italia and the CDS–Bond basis — the alpha is real: positive, robust, positively skewed. Not nickels in front of a steamroller. The iTraxx skew is the informative exception, absorbed by systematic risk factors. And where these arbitrages live isn't set by asset class — it's set by the network of intermediaries behind the market, and ultimately by who holds the bonds.

[Pause.] Thank you.

Thank you for your attention. I would be glad to take your questions.